UTS LAB GENAP

Nama : Silfiana Dwi Nabiilah

NIM : 221344028

Kelas : 2A TNK

Mata Kuliah : Mikroprosesor dan Interface

Dosen : Bapak Ferry Satria, BSEE., M.T.

Ibu Rahmawati Hasanah. S.ST., M.T.

1. Buat program untuk mengisikan 9 data Desimal (100, 85, 75, 60, 50, … dst) ke Blok memori kemudian jumlahkan seluruh data tersebut dan simpan hasilnya pada register CX.

   Jawab:

Program	Hasil dan Kalkulator

```
TITLE UTS NO1
.MODEL SMALL
.STACK 100H
.DATA
DAT1 DW 16 DUP(0)

.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
    MOV SI,0
    ; INISIALISASI 2 DATA PERTAMA
    MOV [DAT1+SI],100H
    INC SI
    INC SI
    MOV [DAT1+SI],85H
    MOV SI,0
    MOV AH,0
    MOV DI,4

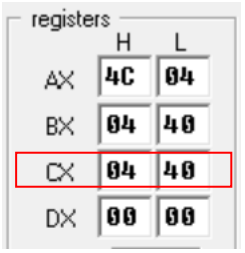
     ; MEMBUAT DERETAN DATA YAITU 100, 85, 75, 60,.. DST
L1:  MOV AX,[DAT1+SI]
     SUB AL,25H
     SBB AH,0
     DAS

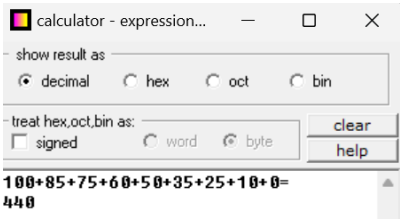
     MOV [DAT1+[SI+4]],AX
     INC DI
     INC SI
     INC SI
     MOV AH,0
     MOV AX,[DAT1+SI]
     SUB AL,25H
     SBB AH,0
     DAS
     MOV [DAT1+[SI+4]],AX
     MOV AH,0
     INC SI
     INC SI
     INC DI

    CMP SI, 9H
    JB L1
    MOV SI,0
    MOV CX,0

    ; LOOP PENJUMLAHAN SEMUA DERET
    MOV BX,[DAT1+11H]
    MOV SI,0FH
L2: MOV AX,BX
    CLC
    ADD AL,BYTE PTR [DAT1+[SI-1]]
    DAA
    ADC BH, 0
    MOV BL, AL
    MOV AL, BH
    ADD AL, BYTE PTR  [DAT1+[SI-2]]
    DAA
    MOV BH,AL
    DEC SI
    DEC SI
    CMP SI,1
    JA L2
    MOV CX,BX
    MOV AH,4CH
    INT 21H

    MAIN ENDP
END MAIN
```





2. Buat program untuk mengubah data Heksadesimal 6 digit menjadi Oktal. Data input dan Output disimpan pada data area.

   Jawab:

Program	Hasil dan Kalkulator

```
TITLE UTS NO2
.MODEL SMALL
.STACK 100H
.DATA
BM1 DB 1AH,15H,1FH
BM2 DW 3 DUP (0)

.CODE
MAIN PROC
      MOV AX,@DATA
      MOV DS,AX
      MOV AX,0
      MOV BX,0
      MOV CX,0
      MOV DX,0
      MOV SI,0
      MOV DI,0

LUP:  MOV CX,0
      MOV BX,0
      MOV CL,[BM1+SI+2]
      MOV BH,[BM1+SI+1]
      MOV BL,[BM1+SI]
      MOV DX,BX
      AND DX,0FFF8H
      ADD DX,BX
      RCL CX,1

      MOV BX,DX
      AND BX,0FF80H
      ADD DX,BX
      RCL CX,1
      MOV BX,DX
      AND DX,0F800H
      ADD DX,BX
      RCL CX,1
      MOV BX,DX
      AND BX,08000H
      ADD DX,BX
      RCL CX,1
      MOV BX,CX
      AND BX,0FFF8H
      ADD CX,BX
      MOV BX,CX
      AND BX,0FF80H
      ADD CX,BX

      MOV BX,CX
      AND BX,8000H

      MOV [BM2+DI+2],CX
      MOV [BM2+DI],DX
      MOV BX,0
      MOV CX,0
      MOV DX,0
      ADD SI,3
      ADD DI,4
      CMP SI,3
      JNZ LUP

      MOV AH,4CH
      INT 21H

MAIN ENDP
END MAIN
```

